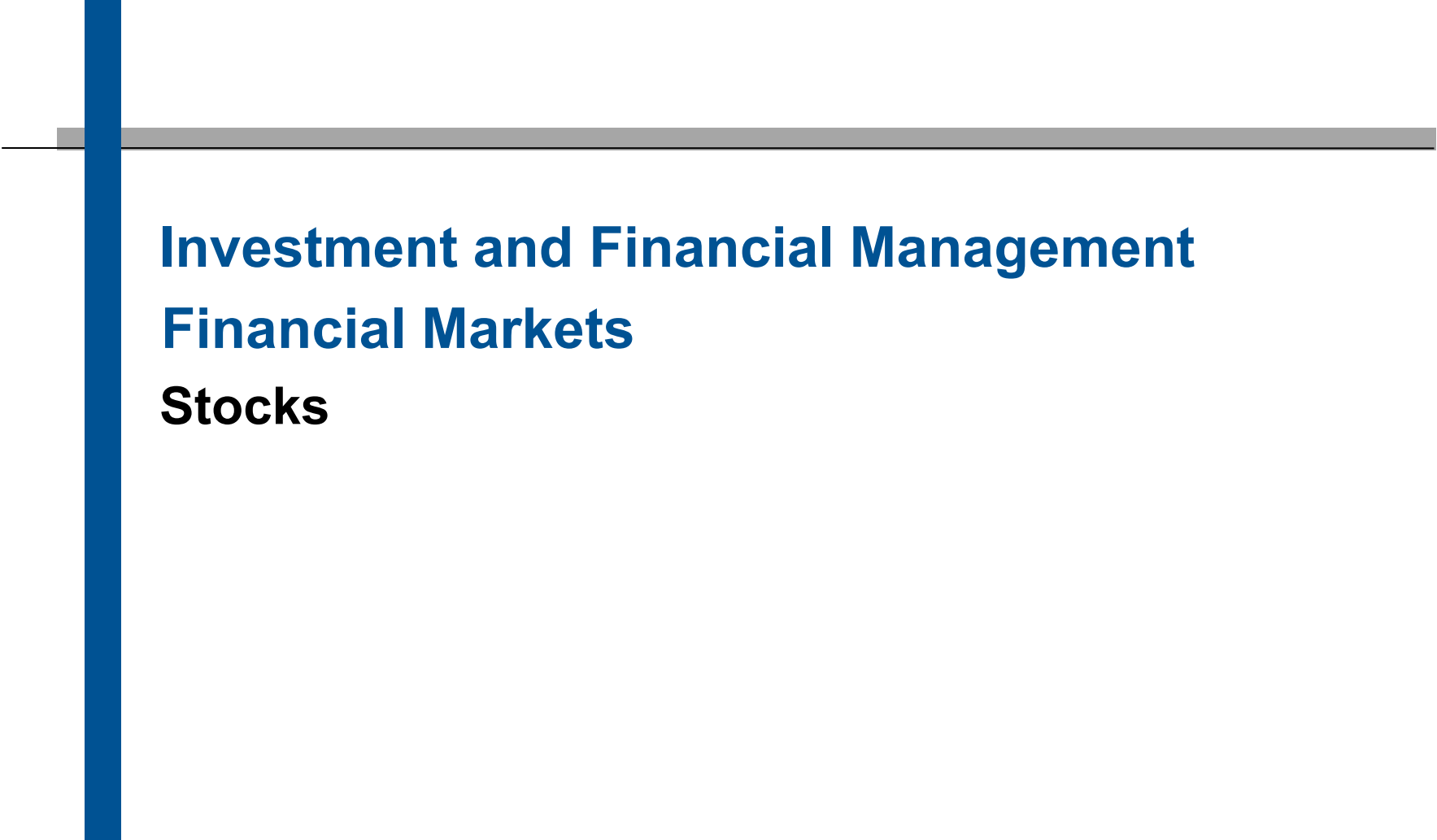




Investment and Financial Management

Financial Markets

Stocks

III. Time-varying growth model (two-phase growth model)

1. Constant growth with factor w until N and then zero-growth

$$(1): P_0 = \underbrace{\sum_{t=1}^N \frac{D_0 * (1 + w)^t}{(1 + r_e)^t}}_{\text{growth } w \text{ (until } N\text{)}} + \underbrace{\frac{D_0 * (1 + w)^N}{r_e} * \frac{1}{(1 + r_e)^N}}_{\text{perpetuity with no growth}}$$

2. Constant growth with factor w_a until N and then constant growth with w_b

$$(1): P_0 = \underbrace{\sum_{t=1}^N \frac{D_0 * (1 + w_a)^t}{(1 + r_e)^t}}_{\text{growth } w_a \text{ (until } N\text{)}} + \underbrace{\frac{D_0 * (1 + w_a)^N * (1 + w_b)}{r_e - w_b} * \frac{1}{(1 + r_e)^N}}_{\text{perpetuity with growth } w_b}$$

(A.5) A stock has paid a dividend of 2 *EUR* in the last year. It is expected that the dividend will grow at a rate of 8% over the next three years. Afterwards, the growth will go down to a permanent rate of 4%. What is the value of the stock if the costs of capital equal 12%?

$$(1): P_0 = \sum_{t=1}^N \frac{D_0 * (1 + w_a)^t}{(1 + r_e)^t} + \frac{D_0 * (1 + w_a)^N * (1 + w_b)}{r_e - w_b} * \frac{1}{(1 + r_e)^N}$$

$$(2): P_0 = \frac{2 * 1.08^1}{1.12^1} + \frac{2 * 1.08^2}{1.12^2} + \frac{2 * 1.08^3}{1.12^3} + \frac{2 * (1 + 0.08)^3 * (1 + 0.04)}{0.12 - 0.04} * \frac{1}{(1 + 0.12)^3}$$

$$(3): P_0 = 28.89 \text{ EUR}$$

(A.6) Analysts predict the dividends for firm A to equal 2.50 *EUR*, 2.75 *EUR*, and 2.96 *EUR*. It is expected that the company will grow after these three years with a constant rate of 4%. The costs of capital equal 12%. The current stock price is 36 *EUR*. Is it a fair price?

$$(1): P_0 = \sum_{t=1}^N \frac{D_0 * (1 + w_a)^t}{(1 + r_e)^t} + \frac{D_0 * (1 + w_a)^N * (1 + w_b)}{r_e - w_b} * \frac{1}{(1 + r_e)^N}$$

$$(2): P_0 = \frac{2.50}{1.12^1} + \frac{2.75}{1.12^2} + \frac{2.96}{1.12^3} + \frac{2.96 * (1 + 0.04)}{0.12 - 0.04} * \frac{1}{(1 + 0.12)^3}$$

$$(3): P_0 = 33.92 \text{ EUR}$$

The stock is currently overvalued.

5.1 Symbols and terminology

5.2 Pricing of stocks

5.2.1 Stock returns

5.2.2 Dividend discount model

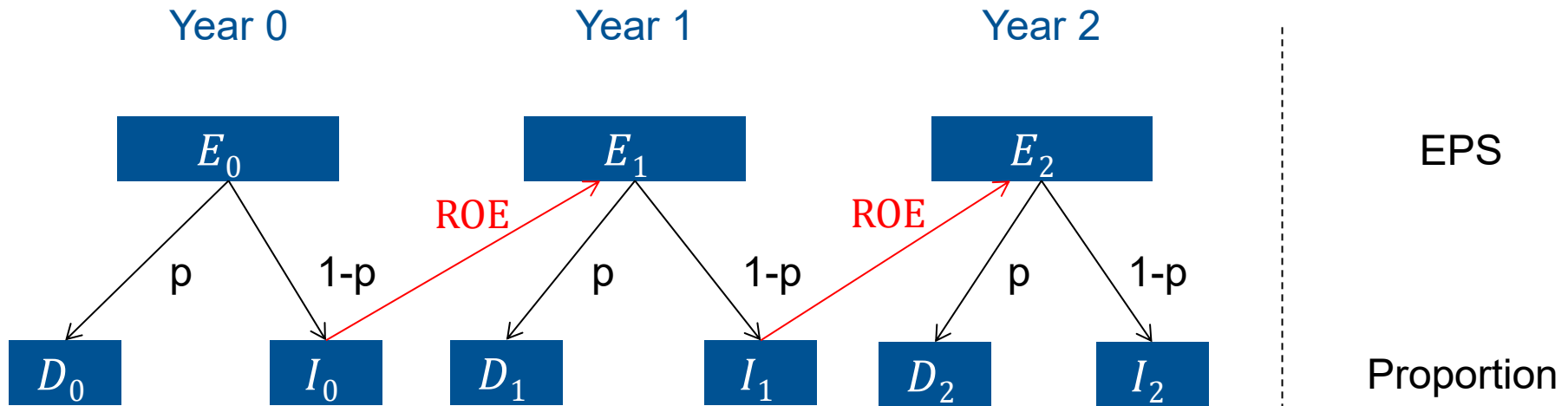
5.2.3 Dividend growth

5.2.4 Price-earnings ratio

5.2.5 Price-book ratio

5.3 Mini case

Idea: Without new investments there is no earnings growth



- I_t as the net investments in year t , D_t as the dividend payment in year t
- p is the dividend ratio, $(1 - p)$ the investment ratio, ROE as the return on equity

$$E_t = E_{t-1} + ROE * I_{t-1} = E_{t-1} + ROE * (1 - p) * E_{t-1}$$

$$E_t = E_{t-1} * (1 + ROE * (1 - p))$$

- w can be predicted as $w = ROE * (1 - p)$
- The growth rate w represents the growth of the earnings per share but also of the dividends per share since, due to the assumptions, p remains constant.

What is the relationship between the dividend growth and the here presented pricing models?

- All projects with $ROE > r_e$ will result a positive net present value
- For young firms which act in a new market, there are a lot of these projects resulting in...
 - a high overall return on equity, and
 - a growth rate w which can be above the costs of equity r_e ($\rightarrow w_a$ in the model with a two-phase growth model)
- Due to the high profitability other firms enter the market and the number of profitable projects goes down. Moreover, as the company gets larger the potential for further growth decreases.
- Then, the firm reaches a constant firm-specific growth rate. The maximum rate equals the overall country-wide growth rate and is smaller than the costs of equity r_e ($\rightarrow w_b$ or w in the model with constant growth).
- Over time (in the model of time-varying growth at the transition from one phase to the other), ROE and w will decrease whereas p will increase.

(A.7) The company Crane Sporting Goods expects earnings per share of 6 *EUR* in the following year. The firm plans to distribute all of their earnings to the shareholders. With this zero-growth strategy the stock price equals 60 *EUR*. Please assume Crane would be able to reduce its dividend ratio to 75% and would use the reserves to open new stores in China. The return on equity equals 12%. What would be the impact of the new strategy on the stock price of the firm?

Scenario I: $D = 6 \text{ EUR} = \text{EPS}$



6 *EUR* to each
Shareholder



0 *EUR* to for new
investments



Scenario II: $D = 75\% * 6 \text{ EUR} = 4.50 \text{ EUR}$



4.50 *EUR* to each
Shareholder



1.50 *EUR* to for new
investments



(A.7) The company Crane Sporting Goods expects earnings per share of 6 *EUR* in the following year. The firm plans to distribute all of their earnings to the shareholders. With this zero-growth strategy the stock price equals 60 *EUR*. Please assume Crane would be able to reduce its dividend ratio to 75% and would use the reserves to open new stores in China. The return on equity equals 12%. What would be the impact of the new strategy on the stock price of the firm?

$$(1): D_1 = 0.75 * 6 \text{ EUR} = 4.50 \text{ EUR}$$

$$(2): w = (1 - p) * ROE = 0.25 * 0.12 = 3\%$$

$$(3): P'_0 = \frac{D_1}{r_e - w} = \frac{4.50}{r_e - 0.03}$$

$$(4): r_e: P_0^{Zero-Growth} = 60 = \frac{D_1}{r_e} = \frac{6}{r_e} \Rightarrow r_e = \frac{6}{60} = 0.10$$

$$(5): P'_0 = \frac{4.50}{0.10 - 0.03} = 64.29 \text{ EUR} > 60 \text{ EUR}$$

$$ROE > r_e \rightarrow P'_0 > P_0^{Zero-Growth}$$

(A.8) Please assume now that the cash reserves from the reduction of the dividend ratio to 75% is used by Crane to finance the opening of new stores in Europe. Here, the return on equity equals only 8%. What would be the effect on the stock price of the company?

Scenario I: $D = 6 \text{ EUR} = \text{EPS}$

Scenario II: $D = 75\% * 6 \text{ EUR} = 4.50 \text{ EUR}$

ROE for new investments → 8%



6 EUR to each Shareholder

0 EUR to for new investments

4.50 EUR to each Shareholder

1.50 EUR to for new investments



(A.8) Please assume now that the cash reserves from the reduction of the dividend ratio to 75% is used by Crane to finance the opening of new stores in Europe. Here, the return on equity equals only 8%. What would be the effect on the stock price of the company?

$$(1): D_1 = 0.75 * 6 \text{ EUR} = 4.50 \text{ EUR}$$

$$(2): w = (1 - p) * ROE = 0.25 * 0.08 = 2\%$$

$$(3): P'_0 = \frac{D_1}{r_e - w} = \frac{4.50}{r_e - 0.02}$$

$$(4): r_e: P_0^{Zero-Growth} = 60 = \frac{D_1}{r_e} = \frac{6}{r_e} \Rightarrow r_e = \frac{6}{60} = 0.10$$

$$(5): P'_0 = \frac{4.50}{0.10 - 0.02} = 56.25 \text{ EUR}$$

$$ROE < r_e \rightarrow P'_0 < P_0^{Zero-Growth}$$

The value of growth

Assumption: If a company distributes all earnings as dividends to the shareholders, it follows a zero-growth strategy since there are no cash reserves to invest. Hence, the current stock price with $p = 1$ and $w = 0$ would represent only the current assets, no future growth potentials.

$$(1): P_0^* = \frac{D_1^*}{r_e} = \frac{EPS_1}{r_e}$$

Difference between the current value of the firm and this zero-growth value represents the present value of future growth potentials **reflecting projects with $ROE > r_e$** which offer returns above the costs of capital.

$$(2): PVGO = P_0 - P_0^* = P_0 - \frac{EPS_1}{r_e}$$

(A.9) The company Crane Sporting Goods expects earnings per share of 6 *EUR* in the following year. The firm plans to distribute all of their earnings to the shareholders. With this zero-growth strategy the stock price equals 60 *EUR*. Please assume Crane would be able to reduce its dividend ratio to 75% and would use the reserves to open new stores in China. The return on equity equals 12%. What would be the present value of this single growth opportunity per share?

$$(1): P_0^{12\%} = \frac{D_1}{r_e - w} = \frac{4.50}{0.1 - 0.03} = 64.29 \text{ EUR}$$

$$(2): P_0^{8\%} = \frac{D_1}{r_e - w} = \frac{4.50}{0.1 - 0.02} = 56.25 \text{ EUR}$$

$$(3): P_0^* = \frac{EPS_1}{r_e} = \frac{6}{0.1} = 60.00 \text{ EUR}$$

$$(4): PVGO^{12\%} = P_0^{12\%} - P_0^* = 64.29 - 60 = 4.29 \text{ EUR}$$

$$(5): PVGO^{8\%} = P_0^{8\%} - P_0^* = 56.25 - 60 = -3.75 \text{ EUR}$$

(A.10) The company Crane Sporting Goods expects earnings per share of 6 *EUR* in the following year. The firm plans to distribute all of their earnings to the shareholders. With this zero-growth strategy the stock price equals 60 *EUR*. Please assume Crane would be able to reduce its dividend ratio to 75% and would use the reserves to open new stores in the USA. The return on equity equals 10%. What would be the impact of the new strategy on the stock price of the firm?

A: none

B: Price goes up

C: Price goes down

$$(1): P_0 = \frac{D}{r_e} \rightarrow r_e = \frac{D}{P_0} = \frac{6}{60} = 10\%$$

$$(2): D_1 = 0.75 * 6 = 4.50$$

$$(3): w = (1 - p) * ROE = 0.25 * 0.10 = 2.5\%$$

$$(4): P_0 = \frac{D_1}{r_e - w} = \frac{4.50}{0.1 - 0.025} = 60.00 = P_0^{Zero-Growth}$$

5.1 Symbols and terminology

5.2 Pricing of stocks

5.2.1 Stock returns

5.2.2 Dividend discount model

5.2.3 Dividend growth

5.2.4 Price-earnings ratio

5.2.5 Price-book ratio

5.3 Mini case

The price-earnings ratio provides information about the relation between the stock price and the annual earnings of a firm

$$\text{Price – earnings ratio} = \frac{\text{share price}}{\text{expected earnings per share}} = \frac{P_0}{EPS_1}$$

$$(1): \frac{P}{E} = \frac{P_0}{EPS_1} = \frac{1}{r_e} + \frac{PVGO}{EPS_1}, \text{ with } P_0 = \frac{EPS_1}{r_e} + PVGO$$

$$(2): \frac{P}{E} = \frac{1}{r_e} * \left(1 + \frac{PVGO}{\frac{EPS_1}{r_e}} \right)$$

The term represents the relation between the value of growth opportunities and the zero-growth value of the firm.

- The **higher** the value of growth relative to the current earnings and the **lower** the costs of equity (as measure of risk), the **higher** is the price-earnings ratio.
- High-growth firms are valued with a higher price-earnings ratio.

(A.11) The ABC AG has a return on equity of 12%, expected earnings per share of 2 EUR, and an expected dividend of 1.50 EUR. The costs of equity equal 10%.

(a) Determine w , P_0 , and the price-earnings ratio by applying the one-phase dividend discount model.

(b) What would change if the dividend ratio would be changed ($p = 0.6$)?

$$(a1): p = \frac{1.50}{2.00} = 0.75 \Rightarrow w = (1 - p) * ROE = 0.25 * 0.12 = 0.03$$

$$(a2): P_0 = \frac{D_1}{r_e - w} = \frac{1.50}{0.10 - 0.03} = 21.43 \text{ EUR}$$

$$(a3): \frac{P}{E} = \frac{P_0}{EPS_1} = \frac{21.43}{2} = 10.71$$

$$(b1): p = 0.6 \Rightarrow D_1 = 1.20 \Rightarrow w = 0.4 * 0.12 = 0.048$$

$$(b2): P_0 = \frac{D_1}{r_e - w} = \frac{1.20}{0.10 - 0.048} = 23.08 \text{ EUR}$$

$$(b3): \frac{P}{E} = \frac{P_0}{EPS_1} = \frac{23.08}{2} = 11.54$$

5.1 Symbols and terminology

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5.2.3 Dividend growth

5.2.4 Price-earnings ratio

5.2.5 Price-book ratio

5.3 Mini case

The price-book ratio provides information about the relation between the market price and the book value of the firm's equity

$$\text{Price} - \text{book ratio} = \frac{\text{share price}}{\text{book value per share}} = \frac{\text{Market value of equity}}{\text{Book value of equity}} = \frac{P_0}{V_e}$$

The price-book ratio and the price-earnings ratio are commonly used to evaluate and classify stocks

- Classification in growth stocks (high price-earnings ratio, high price-book ratio) and value stocks (low price-earnings ratio, low price-book ratio)
- Using ratios in order to compare a firm's stock to its industry peers (see multiple approach)
- Ratios are typically forward looking (next 12 months) as in the example below.

Stock	Index	Industry	$\frac{P}{\bar{E}}$ (2024)	$\frac{P}{\bar{B}}$ (2024)
UBS	SMI	Banks	22.3	1.7
BMW	DAX	Cars	6.3	0.8
Nvidia	Nasdaq	IT	34.8	22.0

(A.12) The ABC AG and the DEF AG have a book value per share of 10 *EUR*, $r_e = 15\%$, and $p = 40\%$. They differ in the return on equity: $ROE_{ABC} = 20\%$ vs. $ROE_{DEF} = 15\%$. Please compute the price-book ratio.

$$(1): EPS_1 = ROE * V_e \Rightarrow D_1 = 0.4 * EPS_1 \Rightarrow w = 0.6 * ROE$$

$$(2): P_0 = \frac{D_1}{r_e - w}; (3): \frac{P}{B} = \frac{P_0}{V_e}$$

$$(ABC1): EPS_1 = 2 \Rightarrow D_1 = 0.8 \Rightarrow w = 0.12$$

$$(ABC2): P_0 = \frac{0.8}{0.15 - 0.12} = 26.67 \text{ EUR}; (ABC3): \frac{P}{B} = \frac{26.67}{10} = 2.67$$

$$(DEF1): EPS_1 = 1.5 \Rightarrow D_1 = 0.6 \Rightarrow w = 0.09$$

$$(DEF2): P_0 = \frac{0.6}{0.15 - 0.09} = 10.00 \text{ EUR}; (DEF3): \frac{P}{B} = \frac{10}{10} = 1.00$$

5.1 Symbols and terminology

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5.3 Mini case

(A.13) Multiple Valuation of Deutsche Bank

Simplified Multiple Valuation of Deutsche Bank Group

Peer Group	P/E 2017	PBV		EPS 2018E	BVpS
Barclays	10.99	0.60		1.89	43.66 €
BBVA	10.70	0.85	Deutsche Bank		
BNP Paribas*	9.09	0.68	Deutsche Bank@Median	22.88 €	38.86 €
Citigroup	10.24	0.82	Deutsche Bank@Mean	24.53 €	40.39 €
Commerzbank*	31.70	0.31	Implied market capitalization	31,753 €	53,933 €
			Share price as of February		
Danske Bank*	11.07	1.25	23th, 2017		18.54 €
			Actual P/E ratio (forward		
HSBC	12.06	0.93	looking)	9.82	
ING	13.89	1.02	Actual PBV	0.42	
JP Morgan Chase	12.18	1.43	Implied ROE	4.32%	
Nordea Bank	12.39	1.40			
Santander*	12.33	0.97			
Société Generale	8.41	0.53			
Unicredit*	14.17	0.50			
Wells Fargo	12.72	1.66			
Mean	13.00	0.93			
Median	12.12	0.89			

Forward P/E ratios taken from Yahoo Finance as of February 24th 2017

* Forward P/E is missing, TTM P/E ratio was used instead